

Question ID: a0da8114

Question

While researching a topic, a student has taken the following notes:

- Tibetan mastiffs are large dogs native to the Himalayas.
- A mutation in their EPAS1 gene prevents excess hemoglobin production.
- A mutation in their HBB gene boosts hemoglobin's oxygen-carrying ability.
- These mutations enable the dogs to withstand hypoxic (low-oxygen) conditions at high altitudes.
- In a 2016 study, Zhen Wang and colleagues noted that Tibetan wolves' DNA has the same EPAS1 and HBB mutations.
- Wang and colleagues determined that the dogs first acquired these mutations by interbreeding with Tibetan wolves around 24,000 years ago.

The student wants to present the conclusion of Zhen Wang and colleagues' 2016 study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

- A. Like Tibetan mastiffs, Tibetan wolves can withstand hypoxic conditions at high altitudes.
- B. Both Tibetan mastiffs and Tibetan wolves have mutations in their EPAS1 and HBB genes, which prevent excess hemoglobin production and boost hemoglobin's oxygen-carrying ability, respectively.
- C. In addition to preventing excess hemoglobin production, a mutation in Tibetan mastiffs' HBB gene boosts hemoglobin's oxygen-carrying ability.
- D. By interbreeding with Tibetan wolves around 24,000 years ago, Tibetan mastiffs acquired the genetic mutations that enable them to withstand hypoxic conditions.